

Sprint 4: Executable Quote Engine

How a safe reference rate becomes an immutable customer offer

In progress

Current status: ADR-009, the quote schema, production migration, and exact domain calculator are delivered. Repository orchestration, PostgreSQL quote evidence tests, private API, and policy activation remain in progress.

Connection to Sprint 3

Sprint 3 stores an accepted reference rate and every provider fact behind it. Sprint 4 will select a fresh snapshot, add the configured spread and purchase fees, calculate exact fiat and crypto amounts, and freeze the result until expiry.

Planned Flow

```
accepted Sprint 3 reference snapshot
      + versioned quote policy
      + total fiat debit
          v
fees -> trade amount -> spread-adjusted rate
          v
immutable executable quote with short expiry
          v
Sendaza displays it; Sprint 6 later reserves inventory
```

Approved Economic Rule

The request supplies total fiat debit. Fixed and percentage fees are deducted before conversion. Percentage fee rounds up to fiat atomic units; destination crypto rounds down to asset atomic units. The source snapshot expiry limits quote expiry. A quote never locks Sendaza money or reserves SLE inventory.

Delivered and Planned Evidence

- Twenty focused calculator tests cover fixed, percentage, combined, and zero fees; fee ceiling; destination floor; display half-even rounding; inclusive limits; invalid policy; tiny amount; and PostgreSQL bigint overflow.

- The complete gate passes 32 suites and 178 tests, strict TypeScript, and ESLint.
- Expiry and configuration-change orchestration remains planned.
- Idempotency and concurrent creation tests remain planned.
- Real PostgreSQL quote atomicity and immutability tests remain planned.
- Private API contract and stable error tests remain planned.

Current Limitations

All eight migrations are applied in test and production, but no quote policy is active. No quote repository, orchestration service, or endpoint exists yet. Production pricing remains inactive, so no executable quote can currently be issued.

Glossary

Total debit: all fiat the customer authorizes. **Trade amount:** total debit minus fees. **Spread:** configured adjustment from reference rate to customer rate. **Atomic unit:** smallest supported unit of a currency or asset. **Executable quote:** immutable short-lived economics that a later purchase may consume.

Started 1 September 2026. Sprint 4 remains in progress.